Systems Architecture

Memory and Registers

Sequential Logic

Just as logic gates can be combined to provide useful circuits, so can flip-flops.
Examples include:

- Memory
- Register
- Shift Registers
- Counters
- State Machines

Memory Cell

- **Static Memory**

  Single D-Type

  D-Types are expensive

- **Dynamic Memory**

  Single Capacitor

  Capacitors *leek* over time

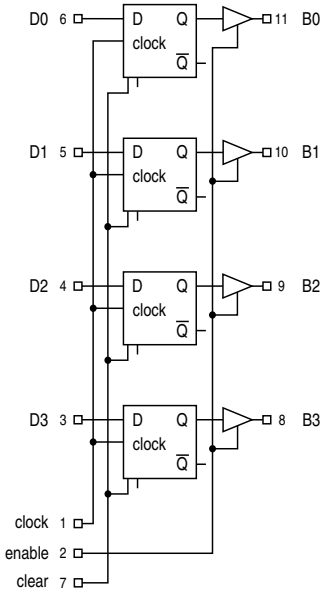
  Refresh to counter leek

  Smaller than D-Types

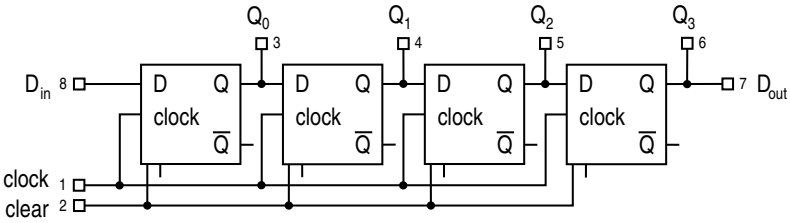
  Cheaper than D-Types

The Register

- Collection of bits (D-Types)
  Byte / Word / Long Word / Extended
  Accessed in parallel
- Write / Load
  Load value when *clocked*
  One device can write at a time
  Input switched to device
- Read / Output Enable
  Output controlled via *enable* signal
  Tri-States used to control output
  Only one device can read at a time
  Output switched to reading device



Shift Register

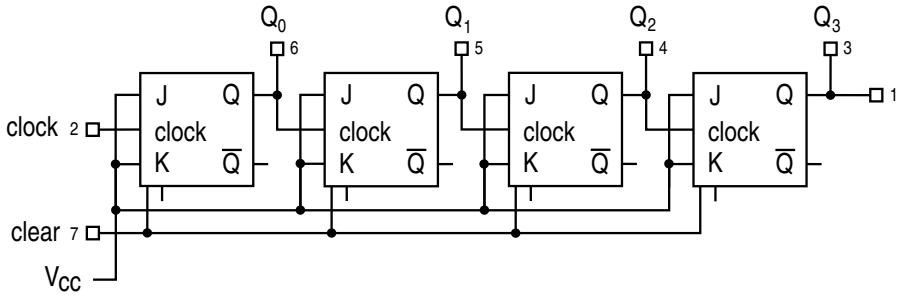


- Move data left / right
- Parallel $\Leftrightarrow$ Serial
- $\times 2$ or $\div 2$

Counters

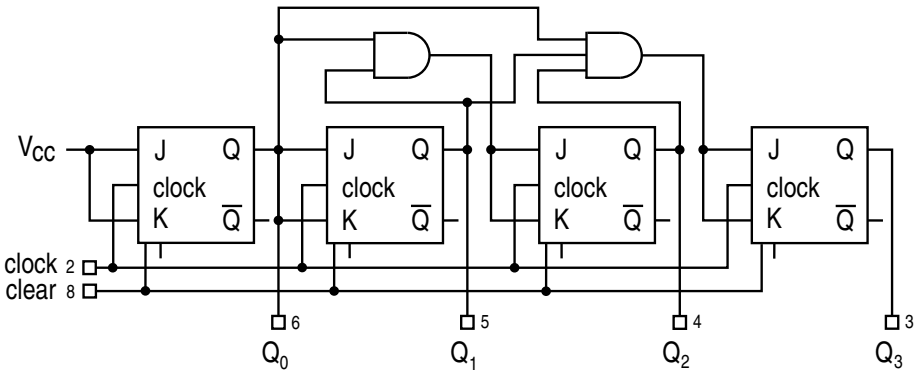
- Single “clock” input
- Counts number of clock pulses
- May have any number of outputs
  n outputs requires n flip-flops
  Can count from 0 to $2^n - 1$
- Cyclic: Loops from end back to start
- up- or down- counting
  Incrementing or Decrementing
- Asynchronous or Synchronous

Asynchronous Counters



- Q used to toggle next JK
- Output ripples through counter
- Stable only at end of inc/dec
- CLR will reset to zero

Synchronous Counters



- All JKs clocked at the same time
- Stable immediately after clocked

Practical Circuits

- What happens at Power On?

  ⇒ Random output

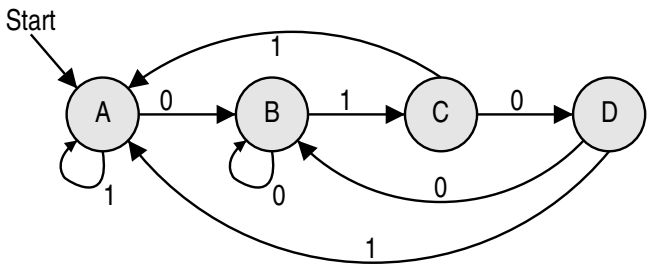
  ⇒ Use preset ($\overline{PR}$) to set to 1

  ⇒ Use clear ($\overline{CLR}$) to set to 0

- Manufacturers provide many pre-built sequential logic circuits as integrated circuits (ICs or chips).

Practical Considerations

- Operating temperature range
- Noise margin
- Propagation delay
- PGA: Programmable Gate Array
- Tradeoffs:
  Weight
  Reliability
  Power Consumption
  Cost
  Size

State Machine

- Graphical tool for working with sequences
- Output dependent on input and current state
- New input changes current state
- Use Flip Flops to hold state